

PixeSci Lean Canvas

Updated strategic positioning | September 2026

Agentic quality control operating system for regulated life-science companies

CORE THESIS

Regulated QC is fragmented across instruments, software, people, procedures, equipment, environmental systems, and external partners. Manual handoffs break continuity between execution and evidence. Small gaps become expensive when they are discovered late. PixeSci operates behind the existing stack to carry context, controls, approvals, and evidence through the workflow as it happens.

FRAGMENTED QC	MANUAL HANDOFFS	EVIDENCE GAPS	EXPENSIVE CONSEQUENCES	PIXESCI ORCHESTRATION	CONTINUOUS QUALITY
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From retrospective reconstruction to prospective, continuous quality.

No new instruments. No rip-and-replace software. One governed execution layer across the Quality Unit.

1 Problem

The problem is not fragmentation alone. The commercial pain is what fragmentation becomes when evidence gaps survive long enough to reach review, release, an audit, or an inspection.

PROBLEM STATEMENT

Regulated QC work is fragmented across systems that were never designed to operate together. Instruments, CDS, LIMS/ELN, QMS, SOPs, spreadsheets, equipment, environmental systems, and external partners each hold part of the workflow. Manual handoffs separate execution from evidence, allowing transcription errors, missing source records, unreconciled repeat testing, missed audit events or equipment conditions, and weak investigation context to pass downstream. Quality is often reconstructed after the work instead of being carried through the work. Existing systems solve individual functions; no operating layer governs the workflow end-to-end.

WHAT BREAKS

- Missing raw records or incomplete source data.
- Source records, summaries, logs, calculations, and instrument histories disagree.
- Aborts, trial runs, reinjections, or repeat testing are not reconciled.
- OOS results and deviations are closed without adequate root-cause or CAPA context.
- Calibration, equipment state, temperature, humidity, or environmental conditions are disconnected from the test.
- Warnings, audit-trail events, user actions, approvals, and exceptions are missed before release.

WHY IT BECOMES EXPENSIVE

- Routine compliance tax: repeated data entry, reconciliation, review, investigation, and audit preparation consume Quality and scientist time.
- Remediation cliff: when gaps are found late, firms may need retrospective reviews, consultant-led assessments, retain testing, re-validation, CAPA, and multi-year oversight.
- The current pitch models two FDA-derived remediation scenarios at \$180K–\$540K base and \$415K–\$1.29M base, before additional OOS review, retain testing, recall, or market-loss costs.
- Those cost figures are PixeSci internal illustrative models based on FDA-required remediation scope; they are not FDA-reported company spend.

BUSINESS AND PATIENT CONSEQUENCES

- Batches can be delayed or held while missing or conflicting evidence is resolved.
- Products can be recalled.
- Import alerts and other enforcement actions can restrict market access.
- FDA remediation can require years of retrospective review across testing, records, audit trails, and investigations.
- At the extreme, quality failures can contribute to patient harm, litigation, loss of trust, or business shutdown.

Lean Canvas problem shorthand: Fragmented QC workflows create expensive quality failures because the evidence needed to prove compliant execution is scattered across disconnected systems and reconstructed too late.

2–5 Product & Differentiation

PixeSci should be positioned as the operating and orchestration layer across systems customers already use — not as a replacement LIMS, QMS, CDS, or instrument stack.

2 SOLUTION

- Connect existing CDS, LIMS, ELN, QMS, instruments, standalone equipment, environmental systems, documents, and partner workflows.
- Resolve execution context: approved method/SOP, sample, workflow version, analyst authorization, training, equipment state, and required approvals.
- Execute authorized cross-system work through agents under deterministic controls and approval gates.
- Monitor instrument events, audit trails, reruns, alarms, environmental conditions, and missing evidence as work happens.
- Route exceptions into Quality workflows and preserve attributable, review-ready execution evidence.
- Enter through one painful workflow; expand across additional systems, instruments, teams, and sites.

3 UNIQUE VALUE PROPOSITION

One governed operating layer across fragmented QC.

- Runs behind the customer's existing software and instruments.
- Carries compliance context, approvals, and evidence through the work instead of reconstructing it afterward.
- Turns quality control from a fragmented, retrospective check into a continuous process embedded in execution.
- Customer-facing line: “Talk to your Lab.™”

4 UNFAIR ADVANTAGE

- Legacy and non-API connectivity, including standalone software and instruments.
- Customer-controlled, on-prem and air-gap-capable deployment.
- Agentic execution with deterministic controls, identity, approval gates, signatures, and evidence capture — not generic AI chat.
- Whole-Quality-Unit scope across laboratory systems, equipment, environmental monitoring, investigations, records, and approvals.
- Compounding connector and workflow library: each deployment can create reusable integration knowledge for the next site.

5 KEY METRICS

- Qualified ICP conversations → pilots → paid deployments.
- Time to first production workflow.
- Deployment hours per customer and monthly support hours per site.
- Active workflows and connected systems/instruments per customer.
- Manual handoffs or transcription steps eliminated.
- Exceptions detected before release or record closure.
- Reduction in evidence-reconstruction, review, and investigation time.
- Pilot-to-paid conversion, ARR/ACV, retention, and expansion.

6–9 Market & Commercial Model

The platform can span the Quality Unit, but the initial sale should be workflow-sized: solve one painful regulated workflow, prove value, then expand.

6 CHANNELS

- Founder-led sales and customer discovery.
- Targeted LinkedIn and email outreach to Quality, QC, lab operations, and regulated manufacturing leaders.
- Warm introductions through biotech, manufacturing, incubator, accelerator, and regional life-science networks.
- Workflow-specific pilots and demos rather than broad platform-first deployments.
- QC, biomanufacturing, lab-informatics, and regulated-automation conferences.
- FDA warning-letter and practical quality education content.
- Partnerships with LIMS/QMS/CDS/instrument vendors, consultants, and system integrators.

7 CUSTOMER SEGMENTS

- Beachhead: emerging U.S. life-science QC sites — regulated manufacturers and testing organizations building or scaling QA/QC operations; not generic biotech R&D.
- Primary users: QC scientists/analysts, QA reviewers, lab managers, and quality-operations personnel.
- Economic buyers: Head/VP of Quality, Head of QC/QA, lab or operations leadership, and founders in smaller companies.
- Buying triggers: GMP readiness, tech transfer, first regulated manufacturing/testing, rising QC volume, fragmented outsourced testing, or a recent quality/data-integrity pain.
- Expansion: CDMOs/CMOs, CROs, contract testing, larger pharma/biotech, medical devices, and multi-site Quality operations.

8 COST STRUCTURE

- Product and agent engineering.
- Connector development, testing, and maintenance.
- Local/on-prem compute and deployment infrastructure.
- Security, validation, auditability, and compliance-readiness work.
- Implementation, workflow mapping, and customer support.
- Sales, customer development, conferences, partnerships, legal, and insurance.
- Critical cost-to-product metrics: deployment hours, support hours, connector reuse, and hardware cost per site.

9 REVENUE STREAMS

- Startup: \$18K/site/year; Quality Core + agentic implementation + integration to 5 scientific software/instruments.
- Additional integration: \$3K.
- Tier 1: \$50K/site/year; up to 10 software environments and expanded orchestration.
- Tier 2: \$120K/site/year; up to 15 software environments, cross-platform orchestration, reusable workflows, and deeper QC/QA controls.
- Enterprise: custom pricing for 15+ environments, multi-team or site-wide standardization, and priority deployment/support.
- Expansion revenue from additional workflows, instruments, connectors, teams, and sites.

Integration surface: quality systems • analytical QC • bioassay • bioprocess/PAT • facility & environmental monitoring.